The documentation and process conversion measures necessary to comply with this revision shall be completed by 15 June 1994

INCH-POUND

MIL-S-19500/117K
15 March 1994
SUPERSEDING
MIL-S-19500/117J
8 September 1992

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON
TYPES 1N962B-1 THROUGH 1N992B-1, 1N962BUR-1 THROUGH 1N992BUR-1,
1N962C-1 THROUGH 1N992C-1, 1N962CUR-1 THROUGH 1N992CUR-1,
1N962D-1 THROUGH 1N992D-1, 1N962DUR-1 THROUGH 1N992DUR-1,
JAN, JANTX, JANTXV, AND JANHC

JANS level (see 6.3)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for 500 milliwatt, silicon, voltage regulator diodes having voltage tolerances of 5 percent, 2 percent, and 1 percent. Three levels of product assurance are provided for each device type as specified in MIL-S-19500. One level of product assurance is provided for die.

1.2 Physical dimensions. See 3.3.

1.3 Maximum ratings. Maximum ratings are as shown in table IV herein and as follows:

P_T = 500 mW (DO-35) at T_L = +50°C, L = .375 (9.53 mm); both ends of case or diode body to heat sink at L = .375 (9.53 mm). (Derate I_Z to 0.0 mA dc at +175°C).

P_T = 500 mW (DO-213AA) at T_{EC} = +125°C (derate to 0 at +175°C).

$-65°C \leq T_{op} \leq +175°C$; $-65°C \leq T_{STG} \leq +175°C$.

1.4 Primary electrical characteristics. Primary electrical characteristic are shown in columns 2, 9, 12, and 14 of table IV herein and as follows:

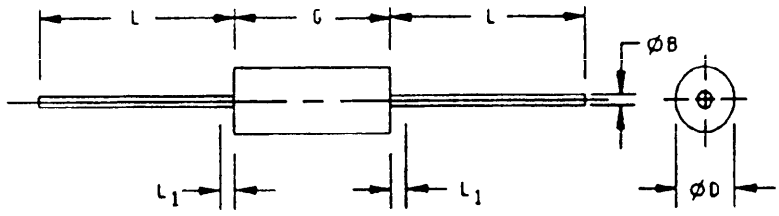
11 V dc $\leq V_Z \leq$ 200 V dc
1N962B-1 through 1N992B-1 are 5 percent voltage tolerance.
1N962C-1 through 1N992C-1 are 2 percent voltage tolerance.
1N962D-1 through 1N992D-1 are 1 percent voltage tolerance.

$R_{\theta JL}$ = 250°C/W (maximum) at L = .375 inch (9.53 mm) (DO-35).
$R_{\theta JEC}$ = 100°C/W (maximum) junction to endcaps (DO-213AA).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ELDT, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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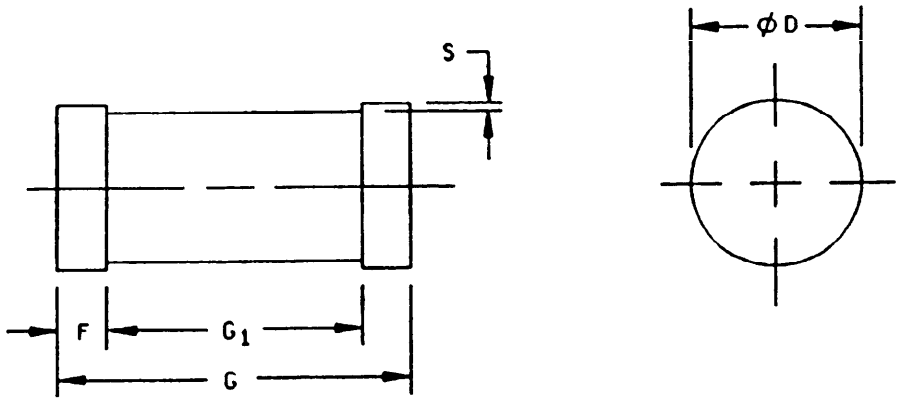


Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ØD	.055	.090	1.40	2.29	3
ØB	.018	.022	0.46	0.56	
G	.120	.200	3.05	5.08	3
L	1.000	1.500	25.40	38.10	
L_1		.050		1.27	4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Package contour optional within ØD and length G. Heat slugs, if any, shall be included within this cylinder but shall not be subject to minimum limit of ØD.
4. Within this zone lead, diameter may vary to allow for lead finishes and irregularities other than heat slugs.

FIGURE 1. Semiconductor device, diode, types 1N962B-1 through 1N992B-1, 1N962C-1 through 1N992C-1, and 1N962D-1 through 1N992D-1 (DO-35).

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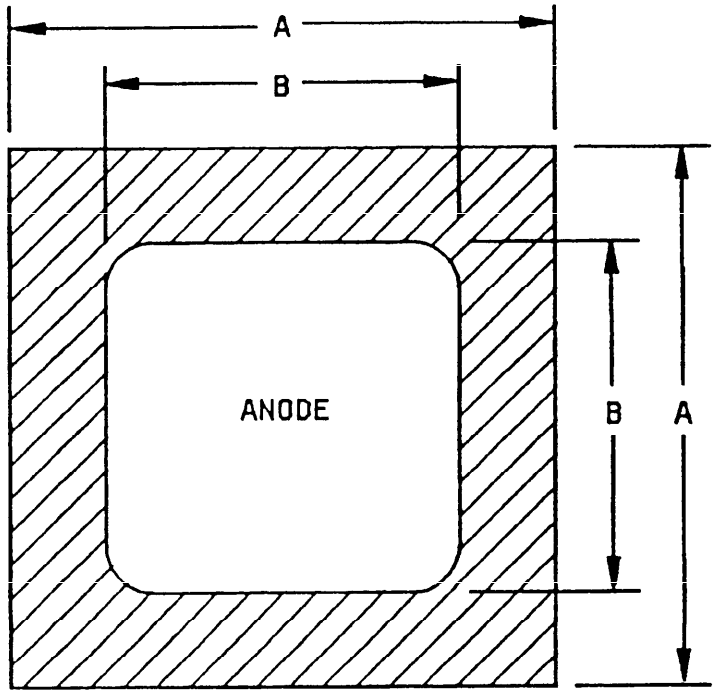


Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
ϕD	.063	.067	1.60	1.70
F	.016	.022	0.41	0.56
G	.130	.146	3.30	3.70
G_1	.100 ref		2.54 ref	
S	.001 min		0.03 min	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Semiconductor device, diode, types 1N962BUR-1 through 1N992BUR-1, 1N962CUR-1 through 1N992CUR-1, and 1N962DUR-1 through 1N992DUR-1 (DO-213AA).

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BACKSIDE IS CATHODE

Ltr	Dimensions							
	JANHCA				JANHCB			
	Inches		Millimeters		Inches		Millimeters	
	Min	Max	Min	Max	Min	Max	Min	Max
A	.021	.025	.53	.63	.024	.028	0.61	0.71
B	.013	.017	.33	.43	.017	.021	0.43	0.53

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die thickness are .010 ±.002 inch (0.25 ±.05 mm). Metallization is: Top (anode) - Al, back (cathode) - Au. Al thickness = 12,000Å minimum for JANHCA and 40,000Å minimum for JANHCB, Au thickness = 3,000Å minimum for JANHCA and 5,000Å for JANHCB.
4. Circuit layout data: For zener operation, cathode must be operated positive with respect to anode.
5. Requirements in accordance with appendix H, are performed in a TO-5 package (see 6.5).

FIGURE 3. Physical dimensions for JANHCA and JANHCB die.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

EC - Endcap.

3.3 Design, construction, and physical dimensions. Design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figures 1 (DO-35), 2 (DO-213AA), and 3 and 4 for JANHC.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-S-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.3.2 Dash one construction. Dash one (-1) devices shall be metallurgically bonded double plug construction in accordance with the requirements of category I, II, or III (see MIL-S-19500, paragraph 30.14 and subparagraphs).

3.4 Marking. Marking shall be in accordance with MIL-S-19500, except at the option of the manufacturer, the following marking may be omitted from the body of the device:

The DO-35 versions may leave off "-" portion of type designation (e.g., JANTX1N962B1).

3.4.1 Marking of UR version devices. For UR version devices only, all marking (except polarity) may be omitted from the body, but shall be retained on the initial container.

3.5 Selection of tight tolerance devices. The C and D suffix devices shall be selected from JAN, JANTX, or JANTXV devices, which have successfully completed all applicable screening, and groups A, B, and C testing as 5 percent tolerance devices. All sublots of C and D suffix devices shall pass group A, subgroup 2, at tightened tolerances. Tighter tolerances for mounting clip temperature shall be maintained for reference purpose to establish correlation. For C and D tolerance levels, T_L = +25°C ±2°C at 0.375 inch (9.53 mm) from body or equivalent.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2.1 JANHC devices. JANHC devices shall be qualified in accordance with appendix H of MIL-S-19500.

4.2.2 Construction verification. Cross sectional photos from three devices shall be submitted in the qualification report.

4.3 Screening (JAN, JANTXV and JANTX levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANTX and JANTXV levels	JAN Level
3a	Temperature cycling	Temperature cycling (in accordance with MIL-S-19500 JTX level)
3c 1/	Thermal impedance (see 4.5.5)	Thermal impedance (see 4.5.5)
9	Not applicable	Not applicable
11	I_R and V_Z	
12	See 4.3.2, t = 48 hours	
13 2/	ΔI_{R1} ≤ 100 percent of initial reading or 50 nA dc, whichever is greater; ΔV_Z ≤ ±2 percent of initial reading, subgroup 2 of table I herein.	

1/ Thermal impedance may be performed any time after sealing provided temperature cycling is performed in accordance with MIL-S-19500, screen 3 prior to this thermal test.

2/ PDA = 5 percent for screen 13, applies to ΔI_{R1}, ΔV_Z and subgroup 2 of table I herein. Thermal impedance ($Z_{\theta JX}$) is not required in screen 13.

4.3.1 Screening (JANHC). Screening of JANHC die shall be in accordance with MIL-S-19500, appendix H. As a minimum, die shall be 100-percent probed in accordance with group A, subgroup 2.

4.3.2 Power burn-in conditions. Power burn-in conditions are as follows: I_Z = column 8 of table IV minimum; mounting and test conditions in accordance with MIL-STD-750, method 1038, test condition B, T_{EC} = +75°C to +125°C for surface mount devices. To better utilize burn-in equipment, higher values of I_Z shall be permitted provided:

a. The junction temperature does not exceed +175°C.

b. The power dissipation does not exceed 500 mW (see figure 6).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps and footnotes of table III herein.

Subgroup	Method	Condition
B2	4066	See 4.5.1
B3	1027	I_Z = column 8 of table IV; (see 4.3.2).
B4	2075	See 4.5.7
B5		Not applicable

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable footnotes and steps of table III herein.

Subgroup	Method	Condition
C2	2036	Tension: Test condition A; weight = 4 pounds; t = 15 seconds. Lead fatigue: Test condition E (tension and lead fatigue are not required for UR-1 suffix devices).
C2	1071	Test condition E.
C3		Not applicable.
C6	1026	I_Z = column 8 of table IV; see 4.3.2.
C7	4071	I_Z = column 5 of table IV; T_1 = +25°C ±5°C; T_2 = T_1 +125°C ±5°C (maximum limit in accordance with column 14 of table IV).

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Surge current (I_{ZSM}). The peak currents shown in column 10 of table IV shall be applied in the reverse direction and these shall be superimposed on the current ($I_Z = I_{Z1}$) (column 5 of table IV) a total of 5 surges at 1 minute intervals. Each individual surge shall be one-half square-wave-pulse of 1/120 of a second duration or an equivalent one-half sinewave with the same effective rms current.

4.5.2 Regulator voltage measurements. The test current shall be applied until thermal equilibrium is attained (90 seconds minimum) prior to reading the breakdown voltage. For this test, the diode shall be suspended by its leads with mounting clips whose inside edge is located at 0.375 inch (9.53 mm) from the body and the mounting clips shall be maintained at a temperature of +25°C +8°C, -2°C. This measurement may be performed after a shorter time following application of the test current than that which provides thermal equilibrium if correlation to stabilized readings can be established to the satisfaction of the Government.

4.5.3 Voltage regulation V_Z (reg). Voltage regulation shall be determined by the difference of the regulator voltage measured at different currents as specified in table I, group A, subgroup 7. Both tests shall be performed at thermal equilibrium. This ΔV_Z shall not exceed column 9 of table IV.

4.5.4 Temperature coefficient of regulator voltage ($_{\alpha}V_Z$). The device shall be temperature stabilized with current applied prior to reading regulator voltage at the specified ambient temperature as specified in paragraph 4.4.3, group C, subgroup 7.

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4.5.5 Thermal impedance ($Z_{\Theta JX}$ measurements). The $Z_{\Theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit for $Z_{\Theta JX}$ in screening (table II of MIL-S-19500) shall be derived by each vendor by means of process control. When three lot date codes have exhibited control, the data from these three lots will be used to establish a fixed screening limit, (not to exceed the group A, subgroup 2 limit). Once a fixed limit has been established, monitor all future sealing lots using a five piece sample from each lot to be plotted on the applicable X, R chart.

a. I_M measurement current - - - - - - - - - - - - - - - - 1 mA - 10 mA.

b. I_H forward heating current - - - - - - - - - - - - - - .5 A - 1.0 A.

c. t_H heating time - 10 ms.

d. t_{MD} measurement delay time - - - - - - - - - - - - - 70 µs maximum.

4.5.5.1 For initial qualification or requalification. Read and record data ($Z_{\Theta JX}$) shall be supplied to the qualifying activity on one lot (random sample of 500 devices minimum). Twenty-two serialized devices shall be sent to the qualifying activity for test correlation.

4.5.6 Thermal resistance. Thermal resistance measurement shall be in accordance with MIL-STD-750, method 3101 or 4081. Forced moving air or draft shall not be permitted across the device during test. The maximum limit for $R_{\Theta JL}$ under these test conditions shall be $R_{\Theta JL}$ (maximum) = 250°C/W, $R_{\Theta JEC}$ (maximum) = 100°C/W. The following conditions shall apply when using method 3101:

a. I_M - 1 mA to 10 mA.

b. I_H - 200 mA to 400 mA.

c. t_H - 25 seconds minimum.

d. t_{MD} - 70 µs maximum.

LS = Lead spacing = .375 inch (9.53 mm) as defined on figure 4 below.
LS = 0 for UR suffix versions

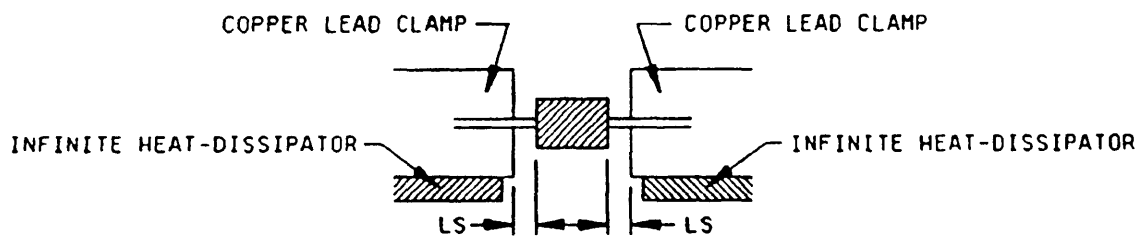


FIGURE 4. Mounting conditions.

4.5.6.1 For initial qualifications and requalifications. Read and record data in accordance with group E herein and shall be included in the qualification report.

4.5.7 Decap internal visual scribe and break. Scratch glass at cavity area with diamond scribe. Carefully snap open. Using 30X magnification, examine the area where die (or bonding material) are in contact with the plugs, verify metallurgical bonding area. If the verification of the metallurgical bonding area is in question, the test method 3101 and test condition limits herein ($Z_{\Theta JX}$) shall be used to determine suitability for use.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	2/ Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Forward voltage	4011	I_F = 200 mA dc	V_F			
1N962-1N985 3/					1.1	V dc
1N986-1N992 3/					1.3	V dc
Reverse current	4016	DC method; V_R = column 11 of table IV	I_{R1}		Column 12	µA dc
Regulator voltage (see 4.5.2)	4022	I_Z = (column 5 of table IV)	V_Z	Column 3	Column 4	V dc
Thermal impedance	3101	See 4.5.5	$Z_{\theta JX}$		35	°C/W
Subgroup 3						
High-temperature operation		T_A = +150°C				
Reverse current	4016	DC method: V_R = column 11 of table IV	I_{R2}		Column 13	µA dc
Subgroup 4						
Small-signal reverse breakdown impedance	4051	I_Z = column 5 of table IV. I_{SIG} = 10 percent of I_Z rms	Z_Z		Column 6	ohms
Small-signal reverse breakdown impedance	4051	I_{ZK} = 250 µA dc I_{SIG} = 25 µA rms	Z_{ZK}		Column 7	ohms
Subgroups 5 and 6						
Not applicable						
Subgroup 7						
Voltage regulation (see 4.5.3)		I_Z = 10 percent of column 8 of table IV (current 1) I_Z = 50 percent of column 8 of table IV (current 2)	ΔV_Z (reg)		Column 9	V dc

1/ For sampling plan, see MIL-S-19500.
2/ Column references are to table IV herein.
3/ Applies to all suffix versions.

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TABLE II. Group E inspection (all product assurance levels).

Inspection 1/	MIL-STD-750		Qualification conformance inspection (sampling plan)
	Method	Conditions	
Subgroup 1			
Temperature cycling	1051	500 cycles	22 devices, c = 0
Electrical measurements		See table III, steps 1, 2, 3, 4, and 5.	
Subgroup 2			
Steady-state dc intermittent life	1037	6,000 cycles I_Z = column 8 of table IV	22 devices, c = 0
Electrical measurements		See table III, steps 2, 3, 4, and 5.	
Subgroup 3			
Not applicable			
Subgroup 4			22 devices, c = 0
Thermal resistance surface mount	3101 or 4081	$R_{\Theta JEC}$ = 100°C/W (maximum) at zero lead length. +25°C ≤ T_{EC} ≤ +35°C (see 4.5.6).	
Thermal resistance leaded	3101 or 4081	$R_{\Theta JL}$ = 250°C/W (maximum) at .375 inch (9.53 mm) lead length +25°C ≤ T_L ≤ +35°C (see 4.5.6).	22 devices, c = 0

1/ A separate sample may be pulled for each test.

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TABLE III. Group A, B, C, and E electrical end-point measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Reverse current	4016	DC method; V_R = column 11 of table IV	I_{R1}		column 12 of table IV	µA dc
2.	Reverse current	4016	DC method; V_R = column 11 of table IV	I_{R3}		2	µA dc
3.	Regulator voltage (see 4.5.2)	4022	I_{Z1} = column 5	V_Z	Column 3	Column 4 of table IV	V dc
4.	Small-signal breakdown impedance	4051	I_{Z1} = column 5 I_{SIG} = 10 percent of I_Z rms	Z_{ZT}		Column 6 of table IV	ohms
5.	Thermal impedance	3101	See 4.5.5	$Z_{\Theta JX}$		35	°C/W

1/ The electrical measurements for table IVb of MIL-S-19500 are as follows:

a. Subgroup 2, see table III herein, steps 1, 3, 4, and 5.

b. Subgroups 3 and 6, see table III herein, steps 2, 3, and 4.

2/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 2, see table III herein, steps 1, 3, 4, and 5.

b. Subgroup 6, see table III herein, steps 2, 3, and 4.

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TABLE IV. Test ratings.

5 percent diodes, types 1N962B-1 through 1N992B-1 and 1N962BUR-1 through 1N992BUR-1													
Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9	Col 10	Col 11	Col 12	Col 13	Col 14
Voltage group	V_Z Nom	V_Z Min	V_Z Max	I_{Z1} Test current	Z_Z Impedance	Z_{ZK} (Knee impedance)	I_{ZM} Max dc current	ΔV_Z (Reg)	I_{ZSM} (Surge)	V_R	I_{R1} at T_A = +25°C V_R = Col 11	I_{R2} at T_A = +150°C V_R = Col 12	αV_Z Temperature coefficient
	Volts	Volts	Volts	mA	ohms	ohms	mA	Volts	mA	Volts	µA	µA	%/°C
1N962B-1	11	10.45	11.55	11.5	9.5	700	35	0.50	590	8.4	1.0	10	+0.073
1N963B-1	12	11.40	12.60	10.5	11.5	700	32	0.55	540	9.1	1.0	10	+0.076
1N964B-1	13	12.35	13.65	9.5	13.0	700	30	0.60	500	9.9	0.5	10	+0.079
1N965B-1	15	14.25	15.75	8.5	16.0	700	26	0.70	433	11	"	10	+0.082
1N966B-1	16	15.20	16.80	7.8	17.0	700	25	0.75	406	12	"	10	+0.083
1N967B-1	18	17.10	18.90	7.0	21	750	21	0.85	361	14	"	10	+0.085
1N968B-1	20	19.0	21.0	6.2	25	750	19	0.95	325	15	"	10	+0.086
1N969B-1	22	20.9	23.1	5.6	29	750	17	1.05	295	17	"	10	+0.087
1N970B-1	24	22.8	25.2	5.2	33	750	16	1.15	271	18	"	10	+0.088
1N971B-1	27	25.7	28.3	4.6	41	750	14	1.30	240	21	"	10	+0.090
1N972B-1	30	28.5	31.5	4.2	49	1000	13	1.45	216	23	"	10	+0.091
1N973B-1	33	31.4	34.6	3.8	58	1000	12	1.60	197	25	"	10	+0.092
1N974B-1	36	34.2	37.8	3.4	70	1000	11	1.75	180	27	"	10	+0.093
1N975B-1	39	37.1	40.9	3.2	80	1000	9.1	1.90	166	30	"	10	+0.094
1N976B-1	43	40.9	45.1	3.0	93	1000	8.8	2.10	151	33	"	10	+0.095
1N977B-1	47	44.7	49.3	2.7	105	1500	7.9	2.25	138	36	"	10	+0.095
1N978B-1	51	48.5	53.5	2.5	125	1500	7.4	2.5	127	39	"	10	+0.096
1N979B-1	56	53.2	58.8	2.2	150	2000	6.9	2.7	116	43	"	10	+0.096
1N980B-1	62	58.9	65.1	2.0	185	2000	6.0	2.9	105	47	"	10	+0.097
1N981B-1	68	64.6	71.4	1.8	230	2000	5.5	3.2	95	52	"	10	+0.097
1N982B-1	75	71.3	78.7	1.7	270	2000	5.1	3.4	86	56	0.5	10	+0.098
1N983B-1	82	77.9	86.1	1.5	330	3000	4.6	3.8	79	62	"	10	+0.098
1N984B-1	91	86.5	95.5	1.4	400	3000	4.2	4.2	71	69	"	10	+0.099
1N985B-1	100	95.0	105.0	1.3	500	3000	3.7	4.4	65	76	"	10	+0.110
1N986B-1	110	104.5	115.5	1.1	750	4000	3.3	4.8	59	84	"	10	+0.110
1N987B-1	120	114.0	126.0	1.0	900	4500	3.1	5.2	54	91	"	10	+0.110
1N988B-1	130	123.5	136.5	0.95	1100	5000	2.7	5.6	50	99	"	10	+0.110
1N989B-1	150	142.5	157.5	0.85	1500	6000	2.4	7.0	43	114	"	10	+0.110
1N990B-1	160	152.0	168.0	0.80	1700	6500	2.2	7.5	40	122	"	10	+0.110
1N991B-1	180	171.0	189.0	0.68	2200	7100	2.0	9.0	36	137	"	10	+0.110
1N992B-1	200	190.0	210.0	0.65	2500	8000	1.8	12.0	32	152	"	10	+0.110

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TABLE IV. Test ratings - Continued.

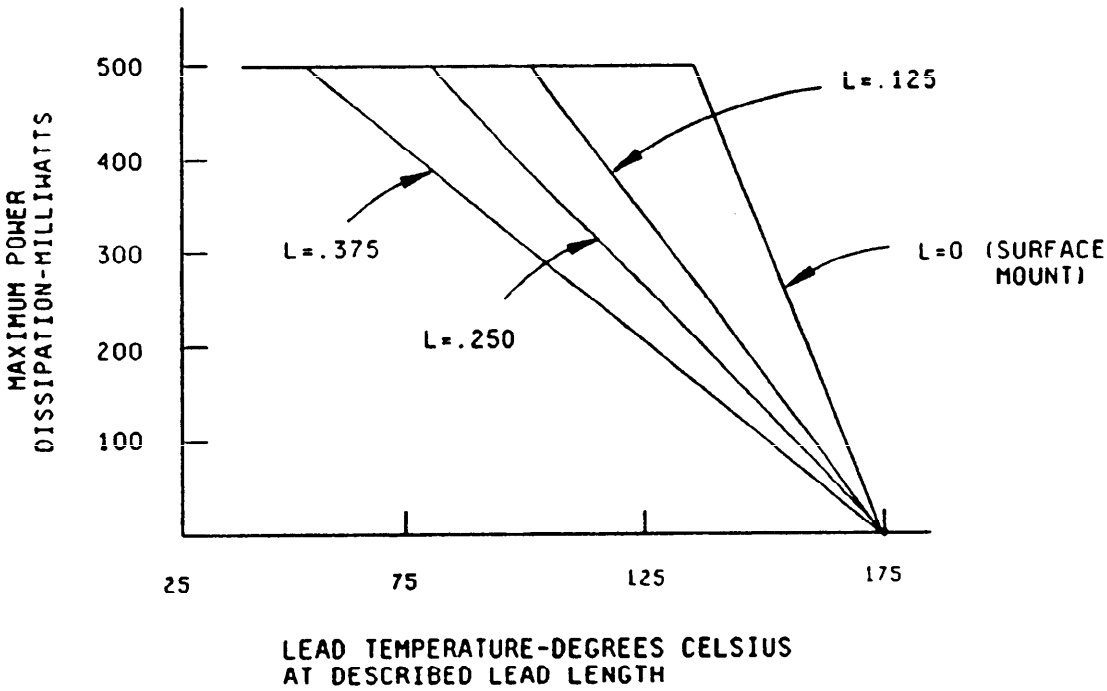
2 percent diodes, types 1N962C-1 through 1N992C-1 and 1N962CUR-1 through 1N992CUR-1													
Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9	Col 10	Col 11	Col 12	Col 13	Col 14
Voltage group	V_Z Nom	V_Z Min	V_Z Max	I_{Z1} Test current	Z_Z Imped-ance	Z_{ZK} (Knee imped-ance)	I_{ZM} Max dc current	ΔV_Z (Reg)	I_{ZSM} (Surge)	V_R	I_{R1} at T_A = +25°C V_R = Col 11	I_{R2} at T_A = +150°C V_R = Col 12	αV_Z Temper-ature coeffi-cient
	Volts	Volts	Volts	mA	ohms	ohms	mA	Volts	mA	Volts	µA	µA	%/°C
1N962C-1	11	10.78	11.22	11.5	9.5	700	35	0.50	590	8.4	1.0	10	+0.073
1N963C-1	12	11.76	12.24	10.5	11.5	700	32	0.55	540	9.1	1.0	10	+0.076
1N964C-1	13	12.74	13.25	9.5	13.0	700	30	0.60	500	9.9	0.5	10	+0.079
1N965C-1	15	14.70	15.30	8.5	16.0	700	26	0.70	433	11	"	10	+0.082
1N966C-1	16	15.68	16.32	7.8	17.0	700	25	0.75	406	12	"	10	+0.083
1N967C-1	18	17.64	18.36	7.0	21	750	21	0.85	361	14	"	10	+0.085
1N968C-1	20	19.60	20.40	6.2	25	750	19	0.95	325	15	"	10	+0.086
1N969C-1	22	21.56	22.44	5.6	29	750	17	1.05	295	17	"	10	+0.087
1N970C-1	24	23.52	24.48	5.2	33	750	16	1.15	271	18	"	10	+0.088
1N971C-1	27	26.46	27.54	4.6	41	750	14	1.30	240	21	"	10	+0.090
1N972C-1	30	29.40	30.60	4.2	49	1000	13	1.45	216	23	"	10	+0.091
1N973C-1	33	32.34	33.66	3.8	58	1000	12	1.60	197	25	"	10	+0.092
1N974C-1	36	35.28	36.72	3.4	70	1000	11	1.75	180	27	"	10	+0.093
1N975C-1	39	38.22	39.78	3.2	80	1000	9.1	1.90	166	30	"	10	+0.094
1N976C-1	43	42.14	43.86	3.0	93	1000	8.8	2.10	151	33	"	10	+0.095
1N977C-1	47	46.06	47.94	2.7	105	1500	7.9	2.25	138	36	"	10	+0.095
1N978C-1	51	49.98	52.02	2.5	125	1500	7.4	2.5	127	39	"	10	+0.096
1N979C-1	56	54.88	57.12	2.2	150	2000	6.9	2.7	116	43	"	10	+0.096
1N980C-1	62	60.76	63.24	2.0	185	2000	6.0	2.9	105	47	"	10	+0.097
1N981C-1	68	66.64	69.36	1.8	230	2000	5.5	3.2	95	52	"	10	+0.097
1N982C-1	75	73.50	76.50	1.7	270	2000	5.1	3.4	86	56	0.5	10	+0.098
1N983C-1	82	80.36	83.64	1.5	330	3000	4.6	3.8	79	62	"	10	+0.098
1N984C-1	91	89.18	92.82	1.4	400	3000	4.2	4.2	71	69	"	10	+0.099
1N985C-1	100	98.0	102.0	1.3	500	3000	3.7	4.4	65	76	"	10	+0.110
1N986C-1	110	107.8	112.2	1.1	750	4000	3.3	4.8	59	84	"	10	+0.110
1N987C-1	120	117.6	122.4	1.0	900	4500	3.1	5.2	54	91	"	10	+0.110
1N988C-1	130	127.4	132.6	0.95	1100	5000	2.7	5.6	50	99	"	10	+0.110
1N989C-1	150	147.0	153.0	0.85	1500	6000	2.4	7.0	43	114	"	10	+0.110
1N990C-1	160	156.8	163.2	0.80	1700	6500	2.2	7.5	40	122	"	10	+0.110
1N991C-1	180	176.4	183.6	0.68	2200	7100	2.0	9.0	36	137	"	10	+0.110
1N992C-1	200	196.0	204.0	0.65	2500	8000	1.8	12.0	32	152	"	10	+0.110

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TABLE IV. Test ratings - Continued.

1 percent diodes, types 1N962D-1 through 1N992D-1 and 1N962DUR-1 through 1N992DUR-1													
Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9	Col 10	Col 11	Col 12	Col 13	Col 14
Voltage group	V_Z Nom	V_Z Min	V_Z Max	I_{Z1} Test current	Z_Z Imped-ance	Z_{ZK} (Knee imped-ance)	I_{ZM} Max dc current	ΔV_Z (Reg)	I_{ZSM} (Surge)	V_R	I_{R1} at T_A = +25°C V_R = Col 11	I_{R2} at T_A = +150°C V_R = Col 12	αV_Z Temper-ature coeffi-cient
	Volts	Volts	Volts	mA	ohms	ohms	mA	Volts	mA	Volts	µA	µA	%/°C
1N962D-1	11	10.89	11.11	11.5	9.5	700	35	0.50	590	8.4	1.0	10	+0.073
1N963D-1	12	11.88	12.12	10.5	11.5	700	32	0.55	540	9.1	1.0	10	+0.076
1N964D-1	13	12.87	13.13	9.5	13.0	700	30	0.60	500	9.9	0.5	10	+0.079
1N965D-1	15	14.85	15.15	8.5	16.0	700	26	0.70	433	11	"	10	+0.082
1N966D-1	16	15.84	16.16	7.8	17.0	700	25	0.75	406	12	"	10	+0.083
1N967D-1	18	17.82	18.18	7.0	21	750	21	0.85	361	14	"	10	+0.085
1N968D-1	20	19.80	20.20	6.2	25	750	19	0.95	325	15	"	10	+0.086
1N969D-1	22	21.78	22.22	5.6	29	750	17	1.05	295	17	"	10	+0.087
1N970D-1	24	23.76	24.24	5.2	33	750	16	1.15	271	18	"	10	+0.088
1N971D-1	27	26.73	27.27	4.6	41	750	14	1.30	240	21	"	10	+0.090
1N972D-1	30	29.70	30.30	4.2	49	1000	13	1.45	216	23	"	10	+0.091
1N973D-1	33	32.67	33.33	3.8	58	1000	12	1.60	197	25	"	10	+0.092
1N974D-1	36	35.64	36.36	3.4	70	1000	11	1.75	180	27	"	10	+0.093
1N975D-1	39	38.61	39.39	3.2	80	1000	9.1	1.90	166	30	"	10	+0.094
1N976D-1	43	42.57	43.43	3.0	93	1000	8.8	2.10	151	33	"	10	+0.095
1N977D-1	47	46.53	47.47	2.7	105	1500	7.9	2.25	138	36	"	10	+0.095
1N978D-1	51	50.49	51.51	2.5	125	1500	7.4	2.5	127	39	"	10	+0.096
1N979D-1	56	55.44	56.56	2.2	150	2000	6.9	2.7	116	43	"	10	+0.096
1N980D-1	62	61.38	62.62	2.0	185	2000	6.0	2.9	105	47	"	10	+0.097
1N981D-1	68	67.32	68.68	1.8	230	2000	5.5	3.2	95	52	"	10	+0.097
1N982D-1	75	74.25	75.75	1.7	270	2000	5.1	3.4	86	56	0.5	10	+0.098
1N983D-1	82	81.18	82.82	1.5	330	3000	4.6	3.8	79	62	"	10	+0.098
1N984D-1	91	90.09	91.91	1.4	400	3000	4.2	4.2	71	69	"	10	+0.099
1N985D-1	100	99.0	101.0	1.3	500	3000	3.7	4.4	65	76	"	10	+0.110
1N986D-1	110	108.9	111.1	1.1	750	4000	3.3	4.8	59	84	"	10	+0.110
1N987D-1	120	118.8	121.2	1.0	900	4500	3.1	5.2	54	91	"	10	+0.110
1N988D-1	130	128.7	131.3	0.95	1100	5000	2.7	5.6	50	99	"	10	+0.110
1N989D-1	150	148.5	151.5	0.85	1500	6000	2.4	7.0	43	114	"	10	+0.110
1N990D-1	160	158.4	161.6	0.80	1700	6500	2.2	7.5	40	122	"	10	+0.110
1N991D-1	180	178.2	181.8	0.68	2200	7100	2.0	9.0	36	137	"	10	+0.110
1N992D-1	200	198.0	202.0	0.65	2500	8000	1.8	12.0	32	152	"	10	+0.110

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Inches	mm
.000	0.00
.125	3.18
.250	6.35
.375	9.53

FIGURE 5. Maximum power versus lead temperature and lead length (see 6.4).

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. For die acquisition, the JANHC letter version should be specified (see figures 3 and 4).

d. Product assurance level and type designation.

6.3 Cross reference substitution. JANS level will no longer be built to this specification. Devices in stock are acceptable provided the date code does not exceed the date of implementation of this specification. Devices required for space flight applications shall meet the requirements of MIL-S-19500/533. Existing supplies of the superseded parts may be used until exhausted. A PIN for PIN replacement table follows, and these devices are directly interchangeable.

JANS superseding PIN	JANS superseded PIN
1N6327	1N964B-1
1N6328	1N965B-1
1N6329	1N966B-1
1N6330	1N967B-1
1N6331	1N968B-1
1N6332	1N969B-1
1N6333	1N970B-1
1N6334	1N971B-1
1N6335	1N972B-1
1N6336	1N973B-1
1N6337	1N974B-1
1N6338	1N975B-1
1N6339	1N976B-1
1N6340	1N977B-1
1N6341	1N978B-1
1N6342	1N979B-1
1N6343	1N980B-1
1N6344	1N981B-1
1N6345	1N982B-1
1N6346	1N983B-1
1N6347	1N984B-1
1N6348	1N985B-1
1N6349	1N986B-1
1N6350	1N987B-1
1N6351	1N988B-1
1N6352	1N989B-1
1N6353	1N990B-1
1N6354	1N991B-1
1N6355	1N992B-1

6.4 Maximum power versus lead temperature. Typical maximum power rating as a function of lead temperature for various lead lengths is shown on figure 5.

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6.5 Suppliers of JANHC die. The qualified JANHC suppliers with the applicable letter version (e.g., JANHCA1N962B) will be identified on the QPL.

JANC ordering information 1/		
PIN	Manufacturer CAGE	
	55801	12954
1N962B	JANHCA1N962B	JANHCB1N962B
1N963B	JANHCA1N963B	JANHCB1N963B
1N964B	JANHCA1N964B	JANHCB1N964B
1N965B	JANHCA1N965B	JANHCB1N965B
1N966B	JANHCA1N966B	JANHCB1N966B
1N967B	JANHCA1N967B	JANHCB1N967B
1N968B	JANHCA1N968B	JANHCB1N968B
1N969B	JANHCA1N969B	JANHCB1N969B
1N970B	JANHCA1N970B	JANHCB1N970B
1N971B	JANHCA1N971B	JANHCB1N971B
1N972B	JANHCA1N972B	JANHCB1N972B
1N973B	JANHCA1N973B	JANHCB1N973B
1N974B	JANHCA1N974B	JANHCB1N974B
1N975B	JANHCA1N975B	JANHCB1N975B
1N976B	JANHCA1N976B	JANHCB1N976B
1N977B	JANHCA1N977B	JANHCB1N977B
1N978B	JANHCA1N978B	JANHCB1N978B
1N979B	JANHCA1N979B	JANHCB1N979B
1N980B	JANHCA1N980B	JANHCB1N980B
1N981B	JANHCA1N981B	JANHCB1N981B
1N982B	JANHCA1N982B	JANHCB1N982B
1N983B	JANHCA1N983B	JANHCB1N983B
1N984B	JANHCA1N984B	JANHCB1N984B
1N985B	JANHCA1N985B	JANHCB1N985B
1N986B	JANHCA1N986B	JANHCB1N986B
1N987B	JANHCA1N987B	JANHCB1N987B
1N988B	JANHCA1N988B	JANHCB1N988B
1N989B	JANHCA1N989B	JANHCB1N989B
1N990B	JANHCA1N990B	JANHCB1N990B
1N991B	JANHCA1N991B	JANHCB1N991B
1N992B	JANHCA1N992B	JANHCB1N992B

1/ Applies to "C" and "D" suffix versions also. Simply replace all "B" suffixes with "C" or "D" as applicable for correct PIN.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, AV, MI, SM
Navy - AS, CG, MC
Air Force - 13, 19, 85, 99

Preparing activity:
DLA - ES

Agent:
DLA - ES

(Project 5961-1630)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/117K	2. DOCUMENT DATE (YYMMDD) 94/03/15

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, DIODE, SILICON, TYPES 1N962B-1 THROUGH 1N992B-1, 1N962BUR-1 THROUGH 1N992BUR-1, 1N962C-1 THROUGH 1N992C-1, 1N962CUR-1 THROUGH 1N992CUR-1, 1N962D-1 THROUGH 1N992D-1, 1N962DUR-1 THROUGH 1N992DUR-1, JAN, JANTX, JANTXV, AND JANC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 513-296-6048	(2) AUTOVON 986-6048
c. ADDRESS (Include Zip Code) Defense Electronics Supply Center ATTN: DESC-ELDT Dayton, OH 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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